Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 10.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. For reasons known only to him, a mathematics professor divides his collection of 1281 teddy bears between his three grandchildren, such that 3 times Alfred's share, 4 times Ben's share and 7 times Charlie's share are all equal.

How many teddy bears does Charlie get?

A 661 **B** 588 **C** 441 **D** 388 **E** 252

2. One fourth of one third of two fifth of a number is 21.

What will be 30% of that number?

A 169 **B** 189 **C** 219 **D** 229 **E** 241

3. A number, when divided by 119 leaves 19 as remainder.

If the same number is divided by 17, what is the remainder?

A 2 **B** 3 **C** 5 **D** 7 **E** 10

4. Rearrange the formula to give c in terms of a and b.

$$a^{\frac{3}{2}} = b^{-\frac{3}{2}}c^{\frac{3}{4}}$$

A $c = \frac{4}{3}a^{\left(\frac{1}{2}\right)}b^{\frac{3}{2}}$ **B** $c = \frac{4}{3}a^{\frac{1}{2}}b^{-\frac{3}{2}}$ **C** $c = a^{\frac{1}{2}}b^{\frac{3}{2}}$ **D** $c = a^{\frac{2}{3}}b^{-2}$ **E** $c = a^2b^2$

5. Let $f(x) = x^2 - 1$ and $g(x) = \sqrt{x} - 1$

Which of the following is equivalent to $\frac{f(x)}{g(x)}$?

A $\left(1 + \sqrt{x}\right)(x - 1)$

B $(1 + \sqrt{x})(1 - x)$

C $\left(1 + \sqrt{x}\right)(1 + x)$

D $(1 - \sqrt{x})(1 - x)$

E $(1 - \sqrt{x})(1 + x)$

6. Let $\sqrt{0.3 + \sqrt{0.08}} = \sqrt{a} + \sqrt{b}$

What is $\frac{1}{a} + \frac{1}{b}$ equal to?

A 25 **B** 20 **C** 15 **D** 10 **E** 5

7. Given $x^2 - 48 = 0$

Solve the equation above.

A $x = \pm 2\sqrt{3}$ **B** $x = \pm 4\sqrt{3}$ **C** $x = \pm 5\sqrt{3}$ **D** $x = \pm 6\sqrt{3}$ **E** $x = \pm 8\sqrt{3}$

8. Given $x^2 - 4x + 3 = 0$

What is the set solution of the equation above?

A $\{1\}$ **B** $\{3\}$ **C** $\{-1, 3\}$ **D** $\{1,3\}$ **E** $\{1, -3\}$

9. Consider the following linear equations:

$$(1)\ kx - 3y = 10$$

$$(2)\ 3x - y = 3$$

Which value of k makes lines (1) and (2) perpendicular?

A 3 **B** $\frac{1}{3}$ **C** $-\frac{1}{3}$ **D** 1 **E** -1

10. Line g and $2x + 5y - 1 = 0$ are two parallel lines.

Line g pass through point $(2, 3)$.

Which of the following is the best equation for line g?

A $2x - 5y = 19$ **B** $-2x + 5y = 19$ **C** $2x + 5y = -4$ **D** $2x + 5y = -2$ **E** $2x + 5y = 19$

11. The lines $ax + 3y - 5 = 0$ and $2x - by - 9 = 0$ intersect at the point $(2, -1)$.

Evaluate $a + b$.

A 6 **B** 7 **C** 8 **D** 9 **E** 10

12. Evaluate the expression.

$$\frac{x^2 - 16}{\sqrt{x - 4}}$$

A $(x + 4)\sqrt{x - 4}$ **B** $(x - 4)\sqrt{x + 4}$ **C** $\sqrt{x + 4}$ **D** $\sqrt{x - 4}$ **E** $x\sqrt{x + 4}$

13. Formula for the curve follows

$$(x-3)^2+(y-1)^2=25$$

Rearrange the formula to give y in term of x.

A $y=1+\sqrt{25-(x-3)^2}$

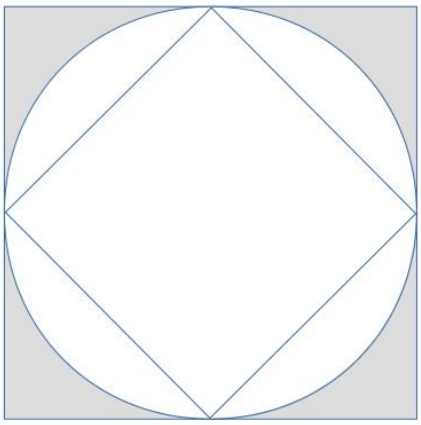
B $y=1-\sqrt{25-(x-3)^2}$

C $y=(\sqrt{25-(x-3)^2})-1$

D $y=\sqrt{24-(x-3)^2}$

E $y=\sqrt{26-(x-3)^2}$

14. The length of each side of the smaller square is $a\sqrt{2}$



What is the best expression for the shaded area?

A $\left(4-\frac{\pi}{4}\right)a^2$ **B** $\left(4-\frac{\pi}{2}\right)a^2$ **C** $(4-\pi)a^2$ **D** $\left(2-\frac{\pi}{2}\right)a^2$ **E** $(2-\pi)a^2$

15. Given $\frac{\sqrt{3}}{\sqrt{6}-\sqrt{2}}$

The simple form of the fraction is …

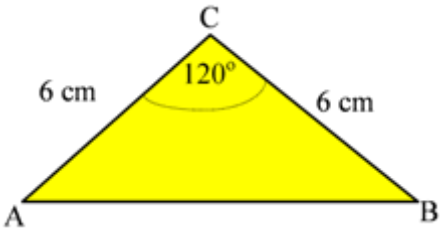
A $4(3\sqrt{2}-\sqrt{6})$ **B** $\frac{1}{4}(3\sqrt{2}+\sqrt{6})$ **C** $\frac{1}{4}(3\sqrt{2}-\sqrt{6})$ **D** $\frac{1}{8}(3\sqrt{2}+\sqrt{6})$ **E** $\frac{1}{8}(3\sqrt{2}-\sqrt{6})$

16. $3\sqrt{24}+2\sqrt{3}(\sqrt{32}-2\sqrt{18})$

Evaluate the above expression.

A $\sqrt{6}$ **B** $2\sqrt{6}$ **C** $4\sqrt{6}$ **D** $6\sqrt{6}$ **E** $8\sqrt{6}$

17. Given the triangle ABC.



The value of AB is

A $6\sqrt{2}$ **B** $6\sqrt{3}$ **C** 9 **D** $9\sqrt{2}$ **E** $9\sqrt{3}$

18. A die is rolled and a coin is tossed.

Find the probability that the die shows an odd number and the coin shows a head.

A 1/8 **B** 1/4 **C** 1/2 **D** 3/4 **E** 1

19. Two six sided dice are rolled.

What is the probability that both dice show odd numbers and their sum is less than 9?

A 1/9 **B** 1/6 **C** 2/9 **D** 1/3 **E** 2/3

20. $\left(\frac{27a^{-5}b^{-3}}{3^5a^{-7}b^{-5}}\right)^{-1}$

Simplify the above expression.

A $(3ab)^2$ **B** $3(ab)^2$ **C** $9(ab)^2$ **D** $\frac{3}{(ab)^2}$ **E** $\frac{9}{(ab)^2}$

21. Let $a = \frac{1}{2}$, $b = 2$, $c = 1$

Evaluate the expression:

$$\frac{a^{-2}bc^3}{ab^2c^{-1}}$$

A 1 **B** 4 **C** 16 **D** 32 **E** 64

22. Imagine a plain, bog-standard clock that reads 5:40.

What is the angle formed between the hour and the minute hands?

A 65° **B** 70° **C** 75° **D** 80° **E** 85°

23. Consider the following equations:

$$y = 9x + 2$$

$$y = 2x + 9$$

What is the solution set of the two equations?

A $(1, 11)$ **B** $(2, 22)$ **C** $(3, 33)$ **D** $(4, 44)$ **E** $(5, 55)$

24. Factorise the above polynomial.

$$3x^3 - 10x^2 - 8x$$

A $x(3x - 4)(x + 2)$

B $x(x - 2)(3x + 4)$

C $x(x + 4)(3x - 2)$

D $x(3x + 2)(x - 4)$

E $x(x - 8)(3x + 1)$

25.

$$f(x) = x^2 - 4$$

$$g(x) = (x + 1)^2$$

Describe the transformation of the graph of $g(x)$ from the graph of $f(x)$.

A Shift 1 unit to the right and then shift 4 units down

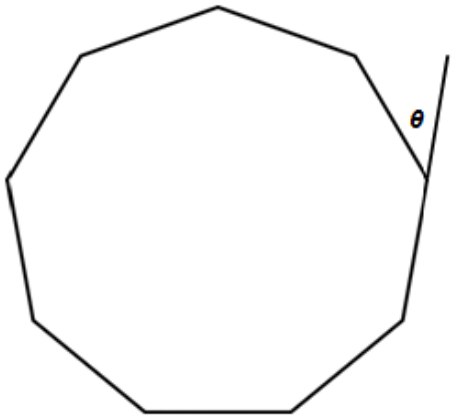
B Shift 1 unit to the left and then shift 4 units up

C Shift 4 units to the right and then shift 1 unit up

D Shift 1 unit to the left and then shift 4 units down

E Shift 1 unit to the right and then shift 4 units up

26. In the regular polygon shown, find the measure of angle θ.



A 30° **B** 35° **C** 40° **D** 45° **E** 50°

27. A teacher asked her class about their pastimes. Apparently, 35 students liked going to the beach, and 25 students liked visiting the local swimming pool. 6 students liked both, and 3 students liked neither.

How many students are there in the class?

A 57 **B** 60 **C** 63 **D** 66 **E** 69

28. A number is decreased by 50% and then increased by 50%.

The resulting number is what percent of the original number?

A 25% **B** 50% **C** 75% **D** 100% **E** 125%

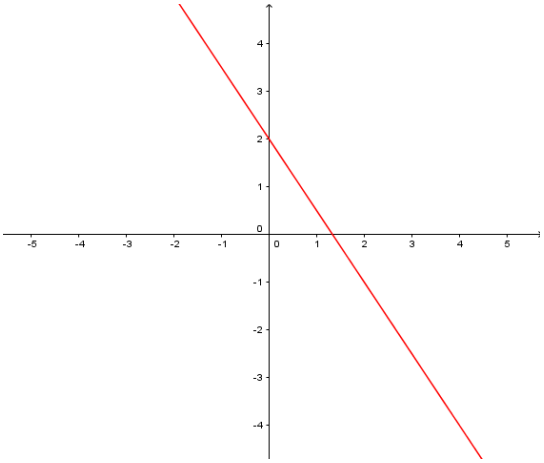
29. Simplify: $2\sqrt{50} - 3\sqrt{27} - 2\sqrt{18} + 3\sqrt{12}$

A $4\sqrt{2} - 3\sqrt{3}$ **B** $3\sqrt{3} - 6\sqrt{2}$ **C** $6\sqrt{2} - 2\sqrt{3}$ **D** $4\sqrt{3} - 6\sqrt{2}$ **E** $3\sqrt{2} - 4\sqrt{3}$

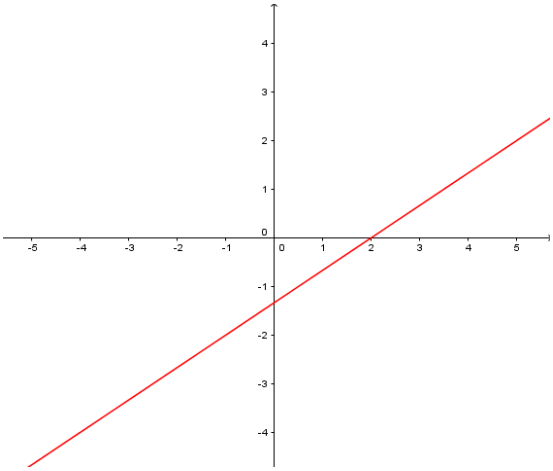
30. $3x - 2y = 4$

Find the graph.

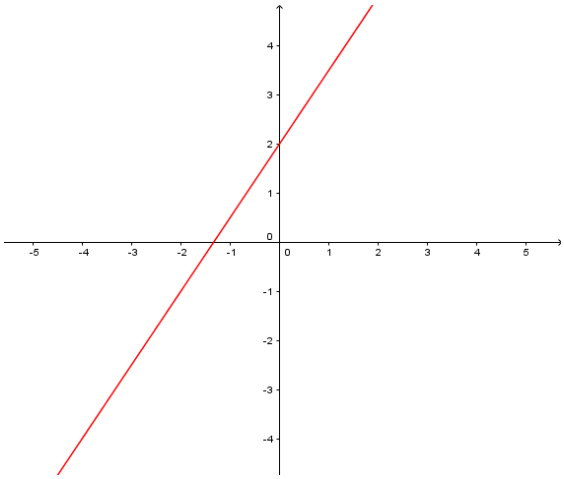
A



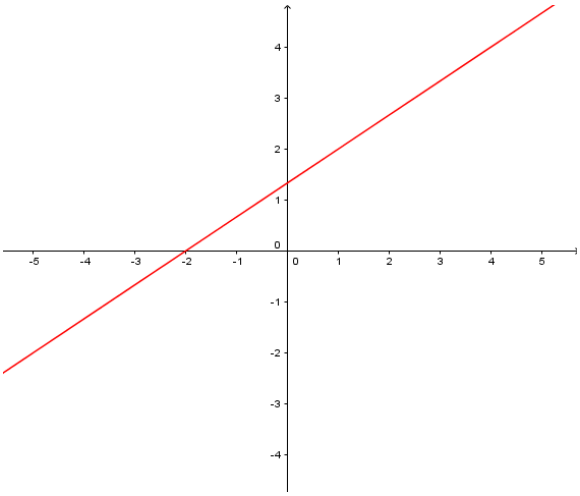
B



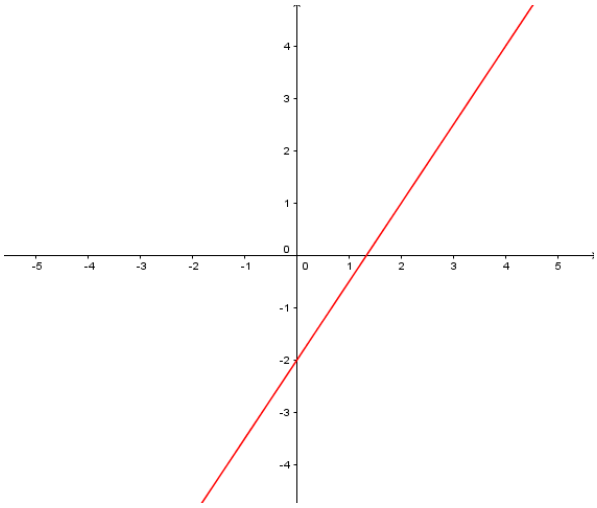
C



D



E



Answer key:

№	Correct Answer	Your answer
1	E	
2	B	
3	A	
4	E	
5	C	
6	C	
7	B	
8	D	
9	E	
10	E	
11	D	
12	A	
13	A	
14	C	
15	B	
16	B	
17	B	
18	B	
19	C	
20	E	
21	B	
22	B	
23	A	
24	D	
25	B	
26	C	
27	A	
28	C	
29	A	
30	E	
		Total: _____